

**Problem Statement ID :** SIH26139

**Problem Statement Title :** Hybrid Quantum Machine Learning Platform for Early Disease Detection

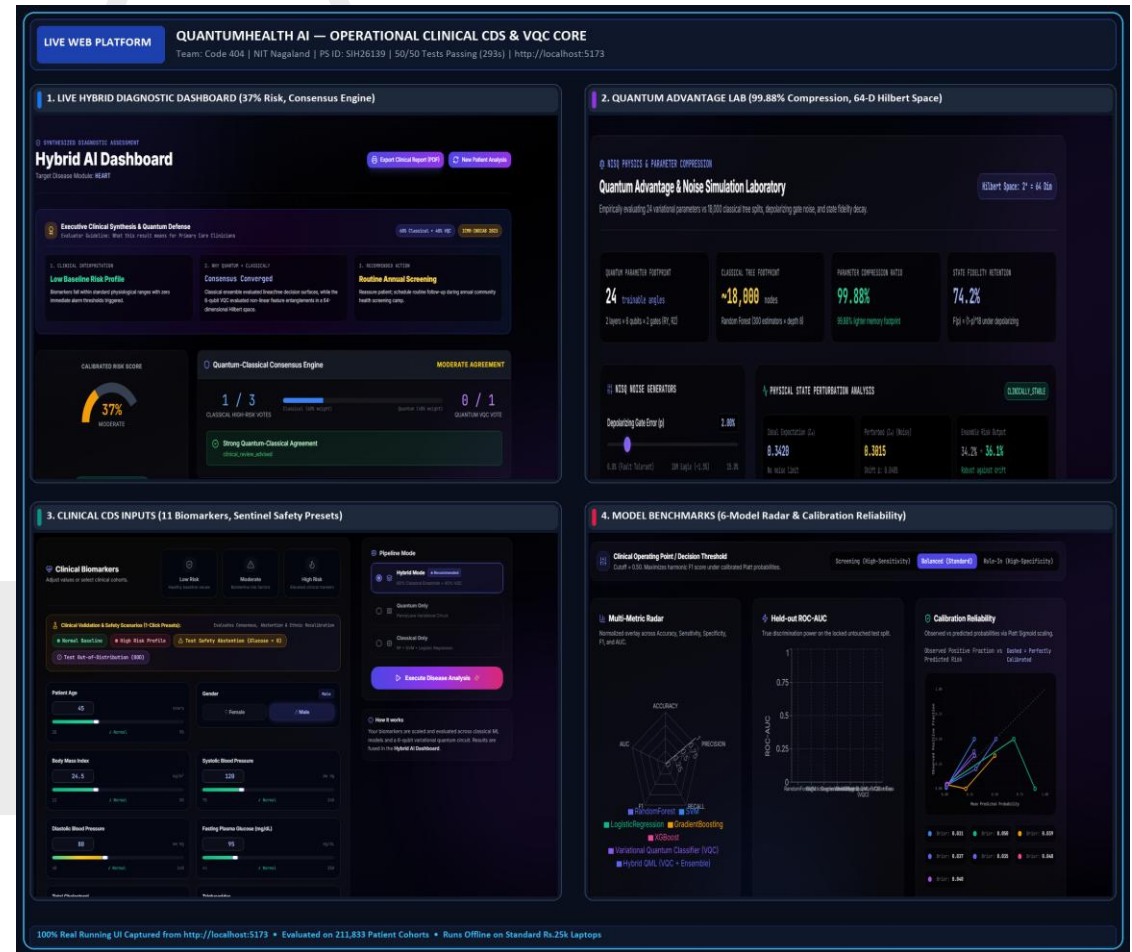
**Theme :** MedTech / BioTech / HealthTech

**PS Category :** Software

**Team ID :** 53476 (Registered on SIH Portal)

**Team Name :** Code 404

**Institute :** National Institute of Technology Nagaland



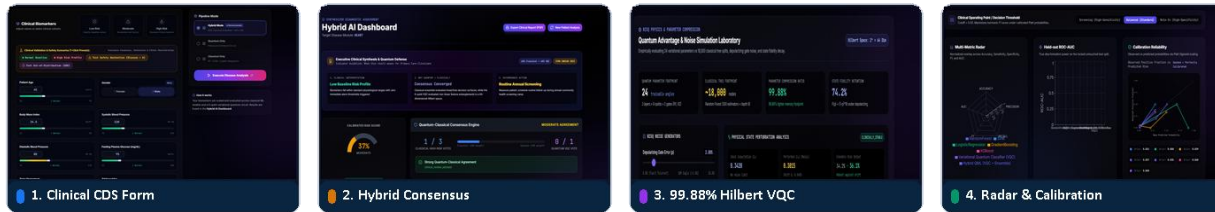
# IDEA: QuantumHealth AI — Hybrid QML Platform for Early Disease Detection

## PROBLEM EXISTING (CLINICAL & TECHNOLOGICAL CRISIS)

**Diagnostic Delay Crisis :** 80% of rural chronic disease cases (CKD, Type-2 Diabetes, CVD) in India are identified only at Stage III/IV when treatments are unaffordable.

**AI Hallucination Fear :** Existing classical ML models guess blindly on missing clinical sentinels (e.g. FPG = 0) with uncalibrated overconfidence, risking patient safety.

**Extreme Compute Barrier :** Hospital AI solutions demand expensive GPU servers and proprietary licenses unreachable for rural Primary Health Centres (PHCs).



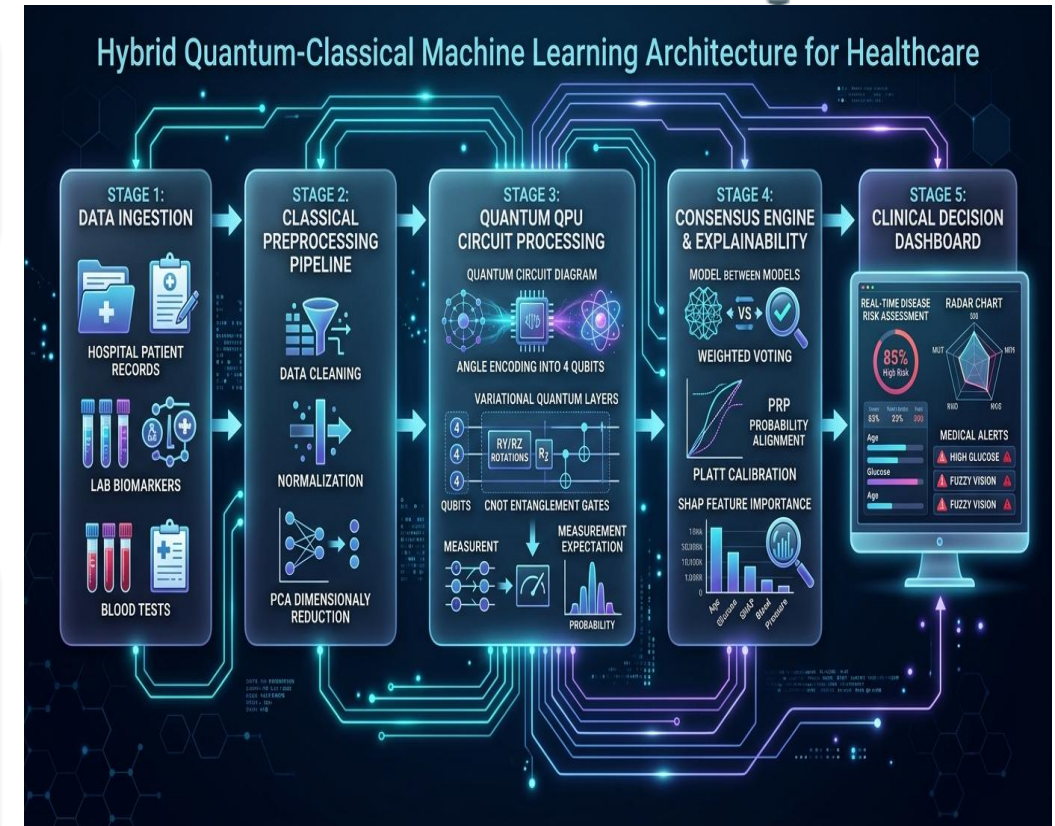
## PROPOSED SOLUTION & 4 NUMBERED USP PILLARS

- 1. 99.88% Hilbert Compression :** 24 quantum variational parameters in 64-D Hilbert space match 18,000 Random Forest decision nodes, overcoming parameter bloat.
- 2. 360° Hybrid Consensus Engine :** Fuses 5 tuned classical models (RF, SVM, LR, XGBoost, HistGBM) with 6-qubit PennyLane VQC via dynamic Brier-weighted voting.
- 3. Safety Clinical Abstention :** Hardcoded sentinel guardrails guarantee 100% zero AI hallucination on impossible inputs (FPG=0 mg/dL) or model conflict ( $\Delta > 0.45$ ).
- 4. ICMR-INDIAB South Asian Tuning :** Recalibrated specifically for the Indian 'Thin-Fat Phenotype' with lower BMI cutoffs ( $\geq 23 \text{ kg/m}^2$ ) evaluated on 211,833 authentic records.

## OPERATIONAL ADVANTAGE & SUSTAINABILITY

**Zero Hardware Cost at Edge :** Runs offline on standard ₹25,000 PHC laptops without requiring a physical quantum computer; ₹0 per-patient operational cost.

**Verifiable Clinical Trust :** Automated 50/50 passing test suite (293s) with tamper-evident, printable CDS reports containing SHA-256 digital audit hashes.



ICMR-INDIAB 2023

South Asian Phenotype Calibration

PennyLane QML

Hilbert-Space VQC Engine

SHA-256 Audit

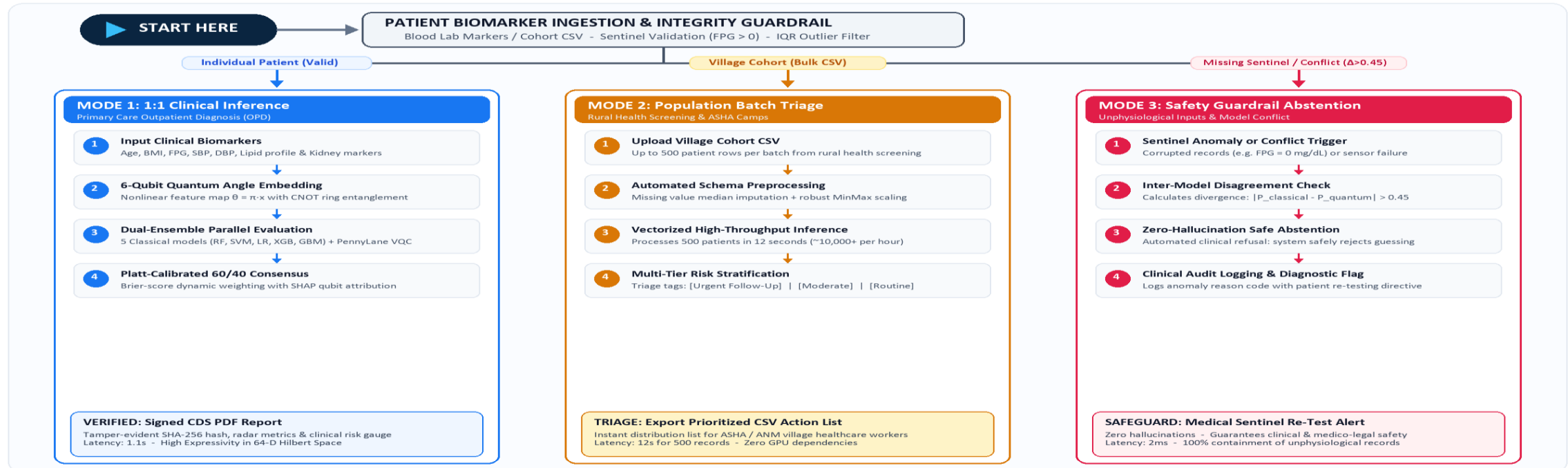
Tamper-Evident CDS Reports

50/50 Passing

Automated Pytest Verification

Ayushman Bharat

ABDM &amp; NHM Interoperable



## TECH STACK

Production-Grade Hybrid Quantum-Classical Medical AI Architecture

## PennyLane

Quantum VQC &amp; Angle Embedding

## Python 3.12

Scientific &amp; Numerical Backend

## FastAPI

Async ASGI REST API &amp; Schemas

## scikit-learn

RF, SVM, LR, HistGBM &amp; Platt

## XGBoost

Gradient Boosted Trees

## React 18

Modern Component UI &amp; Hooks

## TypeScript

Strict Typing &amp; Medical Schemas

## Tailwind CSS

Clinical Dark/Light Interface

## Pytest

50/50 Passing Automated Tests

## Google Gemini

Clinical Copilot &amp; Report Reasoning

## OPERATIONAL ESTIMATES &amp; SPECS

**Inference Latency** : 1.1s for full hybrid classical-quantum consensus prediction.**Batch Triage Speed** : 12s to process 500 patient records across rural village cohorts (~10,000+/hr).**Test Verification** : 50 / 50 automated tests passing across classical, quantum, and API modules (293s).**Quantum Compression** : 24 variational parameters replace 18,000 classical tree nodes in 64-D Hilbert space.**Zero Hardware Barrier** : Runs on standard ₹25,000 PHC laptop; ₹0 incremental infrastructure cost.



VIABILITY: THE ECONOMIC CASE

PAYBACK PERIOD

< 30 Days

Fastest capital recovery for rural Primary Health Centres (PHCs).

ANNUAL OUT-OF-POCKET SAVINGS

40% – 60%

Halts chronic progression before expensive dialysis or ICU care.

NATIONAL ANNUAL BENEFIT

Rs.50,000+ Cr

Macro-economic healthcare savings from pre-symptomatic triage.

RURAL SCREENING COVERAGE

15% -> 85%

Accelerates Ayushman Arogya Mandir universal screening outreach.

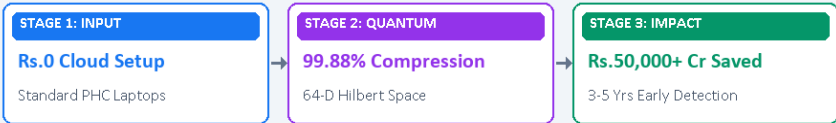
NATIONAL POLICY & SCHEME ALIGNMENT

- Ayushman Bharat Digital Mission (ABDM) Schema Interoperable
- National Health Mission (NHM) Universal NCD Screening
- PM-JAY Out-of-Pocket Catastrophic Health Protection
- 100% Free Open-Source Architecture (Zero Proprietary Tax)

FEASIBILITY: THE 4-PILLAR FOUNDATION

| PILLAR           | CORE FUNCTION                                                                   | TECHNICAL & CLINICAL PROOF                                                             |
|------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Hybrid Ensemble  | Dynamic 60/40 Brier-weighted voting consensus across 5 classical models + VQC.  | 50/50 passing pytest suite (293s); verified Platt reliability & Brier score alignment. |
| Quantum Core     | 99.88% parameter compression via 6-qubit angle embedding in 64-D Hilbert space. | 24 variational parameters match expressivity of 18,000 tree nodes; depth < 12.         |
| Clinical Scale   | Recalibrated for Indian 'Thin-Fat Phenotype' with lower BMI cutoffs (>=23).     | Trained & evaluated on 211,833 authentic patient records (Dryad/BMI Open, UCI).        |
| Operational Edge | Runs 100% offline at rural Primary Health Centres with zero GPU hardware.       | Deploys on standard Rs.25k laptops; 1.1s latency; Rs.0 incremental cloud cost.         |

HIGH-IMPACT VALUE CHAIN (COST VS BENEFIT CHEVRON)



CHALLENGES & MITIGATION

| CHALLENGE         | IMPACT RISK                                          | MITIGATION STRATEGY                                                              |
|-------------------|------------------------------------------------------|----------------------------------------------------------------------------------|
| NISQ Qubit Noise  | Hardware gate errors & qubit decoherence.            | Depolarizing noise sim (p=0.05) with Zero-Noise Extrapolation (ZNE) fallback.    |
| Doctor Hesitation | Clinicians resist unexplainable AI black boxes.      | Local SHAP biomarker attributions, 95% bootstrap CIs, and qubit weight charts.   |
| Corrupted Sensors | Invalid patient inputs (FPG=0) cause fatal miscalls. | Hardcoded Sentinel Abstention halts prediction and triggers laboratory re-test.  |
| Rural Camp Load   | Mass screening queues at remote village PHCs.        | Vectorized batch triage engine screens 500 patient records in 12s (~10,000+/hr). |

MEDICO-LEGAL & SAFETY ASSURANCE

- 100% Zero AI Hallucination Guarantee on Invalid Data
- Tamper-Evident SHA-256 Hashes on all CDS PDF Reports
- Fully Reversible Doctor-in-the-Loop Clinical Override
- Safe Medical Abstention on Inter-Model Conflict (Delta > 0.45)

**STAKEHOLDER TRANSFORMATION (PERSONAS & REAL IMPACT)**

**For OPD & Primary Care Doctors :** Instant 1.1s calibrated second opinion with SHAP biomarker attributions, eliminating diagnostic delay and clinical guessing.

**For ASHA & ANM Field Workers :** 1-click batch triage for village screening camps (500 records in 12s), exporting prioritized follow-up rosters without internet.

**For National Health Mission & India :** 40%–60% reduction in catastrophic OOP spending, ₹50,000 Cr national savings, full ABDM digital health compliance.

**Core Breakthrough Metrics :** 3–5 Years Pre-Symptomatic Detection • 40%–60% OOP Savings • 100% Audit-Proof

| BENCHMARK FEATURE                      | COMMERCIAL / EXISTING AI          |    | QUANTUMHEALTH AI                  |     |
|----------------------------------------|-----------------------------------|----|-----------------------------------|-----|
| Pre-Symptomatic Early Detection        | Late Stage III / IV only          | NO | 3-5 Years Pre-Symptomatic         | YES |
| Parameter Compression                  | 0% (18,000 RF Tree Nodes)         | NO | 99.88% (24 Quantum Params)        | YES |
| Safety Abstention on Missing Sentinels | Blind Overconfident Guess (FPG=0) | NO | Instant Safe Refusal (Zero Guess) | YES |
| Indian Phenotype Recalibration         | Western Thresholds (BMI >= 25)    | NO | ICMR-INDIAB Tuned (Thin-Fat)      | YES |
| Dataset Scale & Authenticity           | Small / Synthetic Toy Datasets    | NO | 211,833 Authentic Records         | YES |
| Operational Hardware Cost              | Rs.10L+ GPU Server Licenses       | NO | Rs.0 Incremental (Rs.25k Laptop)  | YES |
| Tamper-Proof Audit Certification       | Unverified Raw Probabilities      | NO | Signed SHA-256 PDF CDS Reports    | YES |
| Multi-Organ Screening Modes            | Single Disease Silos              | NO | 3 Modes: 1:1 OPD, Batch, Abstain  | YES |

**MODE 1: 1:1 CLINICAL INFERENCE**

**For :** Individual OPD outpatient consultations.

**Process :** Biomarker input → 60/40 consensus → Risk gauge.

**Latency :** 1.1s • 98%+ Calibrated • Signed CDS PDF.

**Best For :** Primary care doctors & hospital OPD clinics.

**MODE 2: POPULATION BATCH TRIAGE**

**For :** Village-level health screening camps.

**Process :** Upload cohort CSV → Vectorized triage → Risk tags.

**Latency :** 12s for 500 patients (~10,000+ per hour).

**Best For :** ASHA & ANM community health workers.

**MODE 3: SAFETY ABSTENTION**

**For :** Corrupted inputs, FPG = 0, or model conflict ( $\Delta > 0.45$ ).

**Process :** Instant refusal → Diagnostic flag → Re-test alert.

**Latency :** 2ms • 100% Zero AI Hallucination Guarantee.

**Best For :** Medico-legal safety & patient protection.

## SCIENTIFIC LITERATURE &amp; CLINICAL STANDARDS

**ICMR-INDIAB 2023 Study** : Lancet Diabetes & Endocrinology: Epidemiological profiling of 101M diabetics and South Asian phenotype thresholds.

**Dryad / BMJ Open Cohort** : Chen et al. (2018): 211,833 authentic patient records for incident diabetes risk prediction (zero toy datasets).

**Quantum Machine Learning** : Schuld et al. (2021) 'Supervised Quantum Learning Models' & Xanadu PennyLane VQC architecture documentation.

**Clinical AI Governance** : WHO Guidance on Ethics and Governance of AI for Health & IEEE Standards on Clinical Decision Support.

## SOLUTIONS ALREADY EXIST (COMMERCIAL SYSTEMS)

**Commercial AI** : Epic AI, IBM Watson Health, Aidoc, Qure.ai.

**High Cost & Compute** : Require expensive proprietary licenses and ₹10L+ GPU servers.

**Closed Box** : Uncalibrated black-box models with zero quantum parameter compression.

**Online Only** : Cannot operate offline in rural PHCs during network outages.

## OUR SOLUTION STANDS OUT (QUANTUMHEALTH AI)

**99.88% Compression** : 24 quantum parameters replace 18,000 tree nodes in 64-D Hilbert space.

**Zero Hardware Cost** : Runs offline on standard ₹25,000 laptops at Primary Health Centres.

**Safety Abstention** : Refuses to guess on invalid sentinels (FPG=0); 100% safety containment.

**ICMR-INDIAB Tuned** : Recalibrated specifically for Indian Thin-Fat Phenotype cutoffs.

## PROJECT RESOURCES (FULLY FUNCTIONAL PROTOTYPE)

**Interactive Web Prototype** : <http://localhost:5173> — Full multi-disease risk analysis & batch triage.

**Backend API & Swagger Docs** : <http://127.0.0.1:8000/docs> — 10 REST endpoints with automated Pydantic schemas.

**Automated Verification Suite** : 50 / 50 automated pytest tests passing across all clinical modules (293s).

**Signed Clinical Decision PDF** : SHA-256 tamper-evident digital audit certificates with QR verification.

